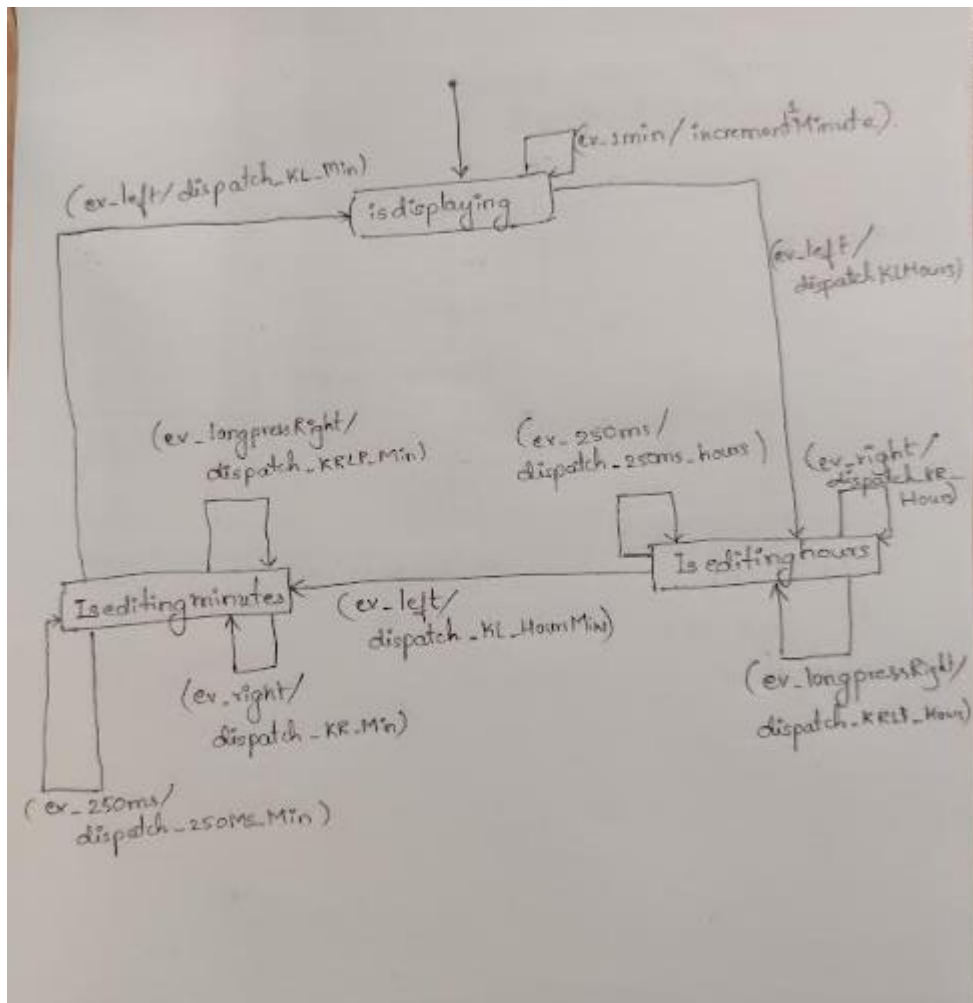
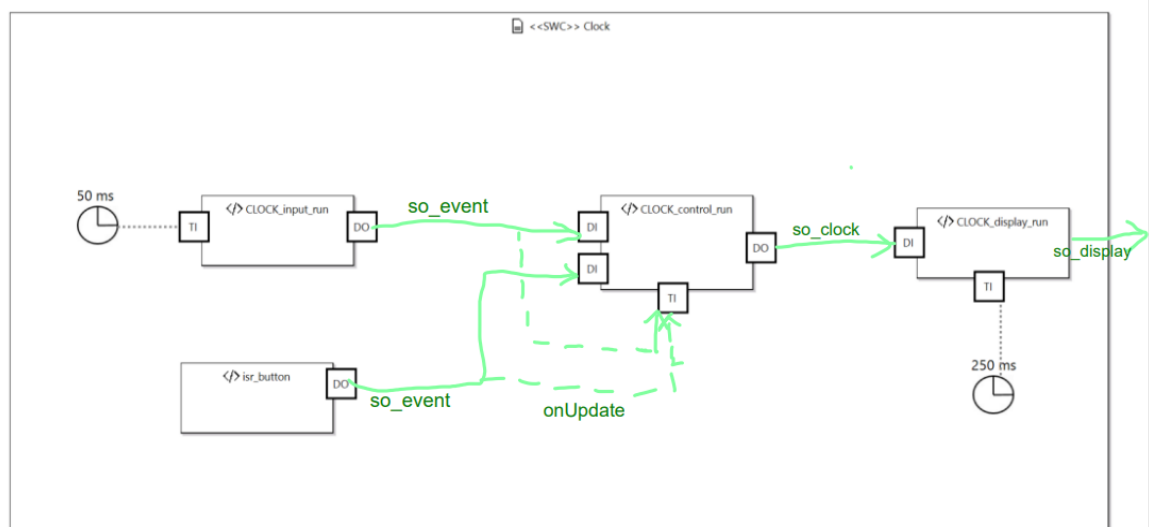


EAA Digital Clock : Assignment 2:

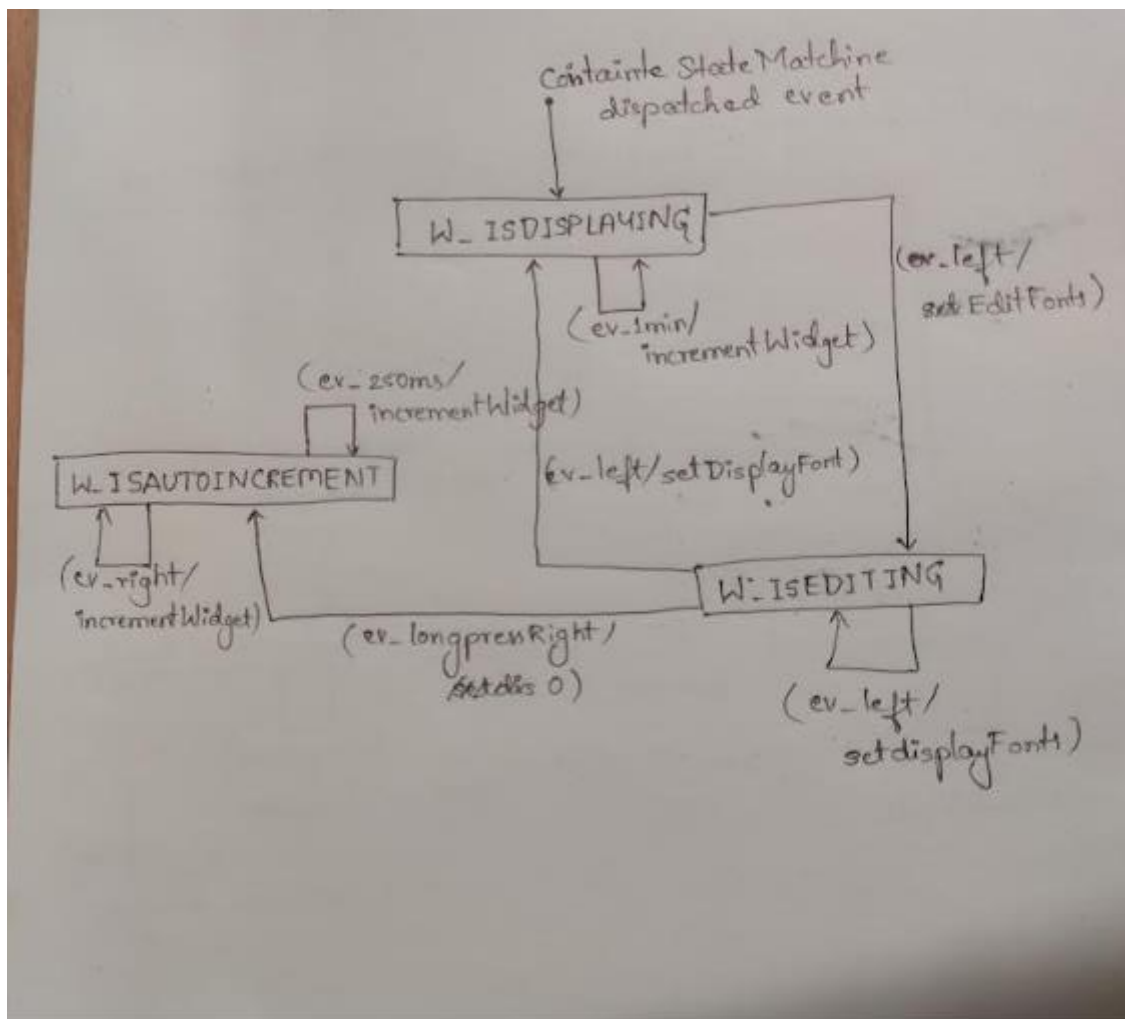
Container State Machine:



SIGNAL FLOW DIAGRAM:



Widget State Machine:



Writing to the same global signal `so_event` from 2 different contexts may result in a loss of data / race – even when you have added resources around the setter / getter functions. Explain why!

There is a high possibility of the `so_event` signal being overwritten by `so_event` by `isr_button` while `so_event` from `CLOCK_input_run` is being copied into `so_clock` or vice versa. In order to avoid this race around condition, we lock the resource until current `so_event` is copied into `so_clock`.

How could you improve the signal flow above to avoid this possible race?

- > use resource to avoid overwriting
- > make `CLOCK_input_run` a cyclic task of 10ms and use polling approach and remove the `ISR_button`